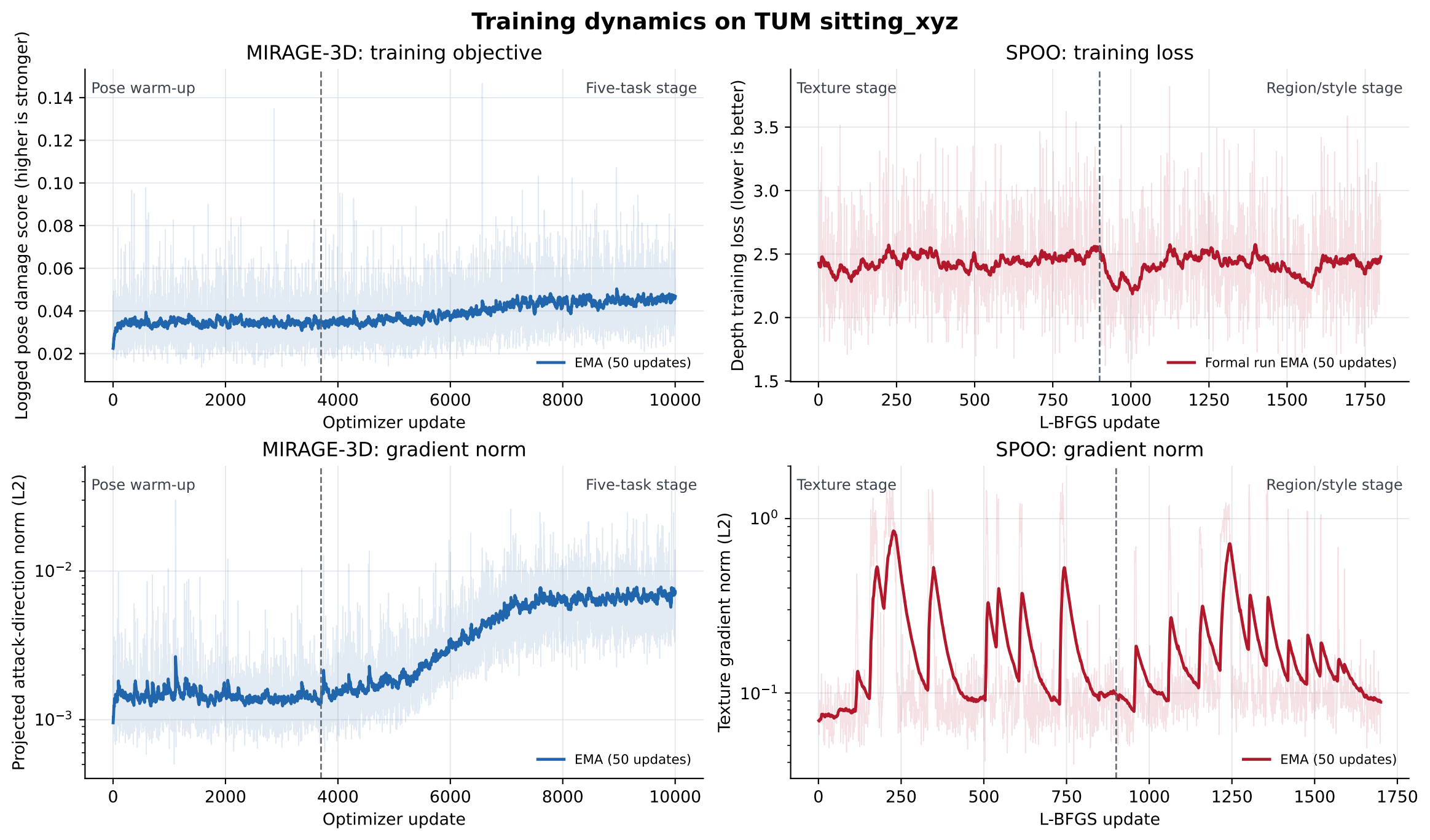

Training dynamics on TUM sitting_xyz
MIRAGE-3D: training objective
Pose warm-up
Five-task stage
Logged pose damage score (higher is stronger)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
EMA (50 updates)
0
2000
4000
6000
8000
10000
Optimizer update
SPOO: training loss
Texture stage
Region/style stage
Depth training loss (lower is better)
3.5
3.0
2.5
2.0
1.5
Formal run EMA (50 updates)
0
250
500
750
1000
1250
1500
1750
L-BFGS update
MIRAGE-3D: gradient norm
Pose warm-up
Five-task stage
Projected attack-direction norm (L2)
10^{-2}
10^{-3}
EMA (50 updates)
0
2000
4000
6000
8000
10000
Optimizer update
SPOO: gradient norm
Texture stage
Region/style stage
Texture gradient norm (L2)
10^{0}
10^{-1}
EMA (50 updates)
0
250
500
750
1000
1250
1500
1750
L-BFGS update